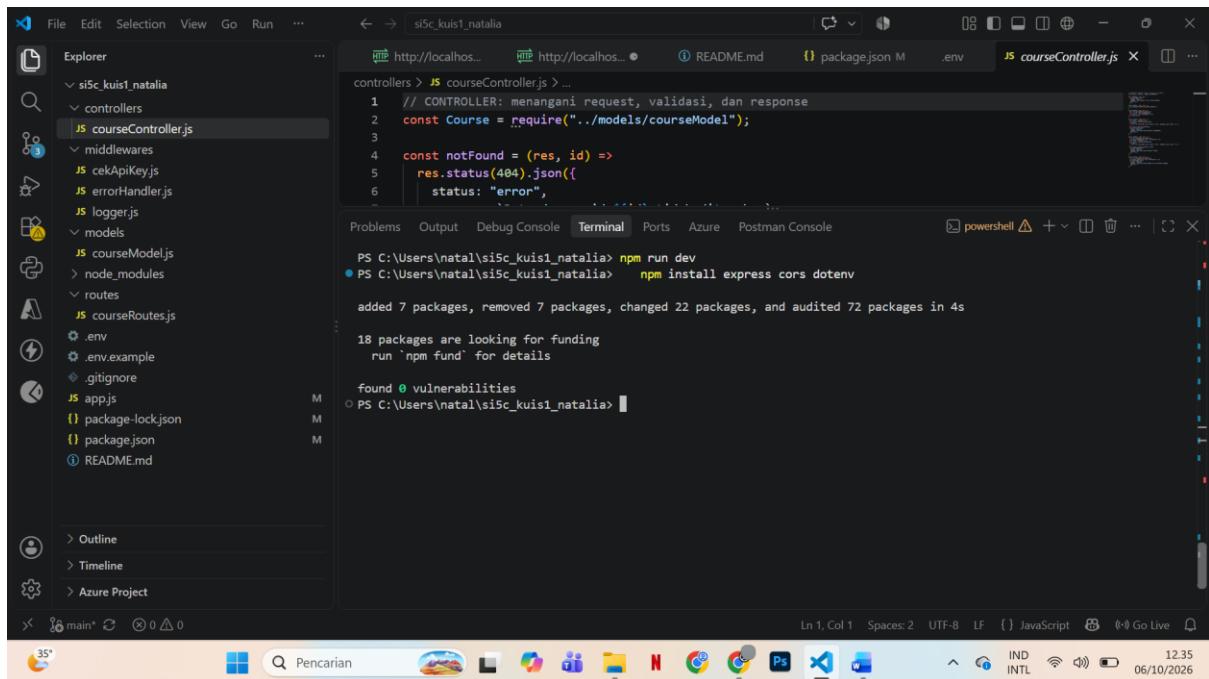


A. Membuat Projek dan Menginstall paket

- mkdir si5c_kuis1_natalia
- cd si5c_kuis1_natalia
- npm init -y
- npm install express cors dotenv



```
1 // CONTROLLER: menangani request, validasi, dan response
2 const Course = require("../models/courseModel");
3
4 const notFound = (res, id) =>
5   res.status(404).json({
6     status: "error",
```

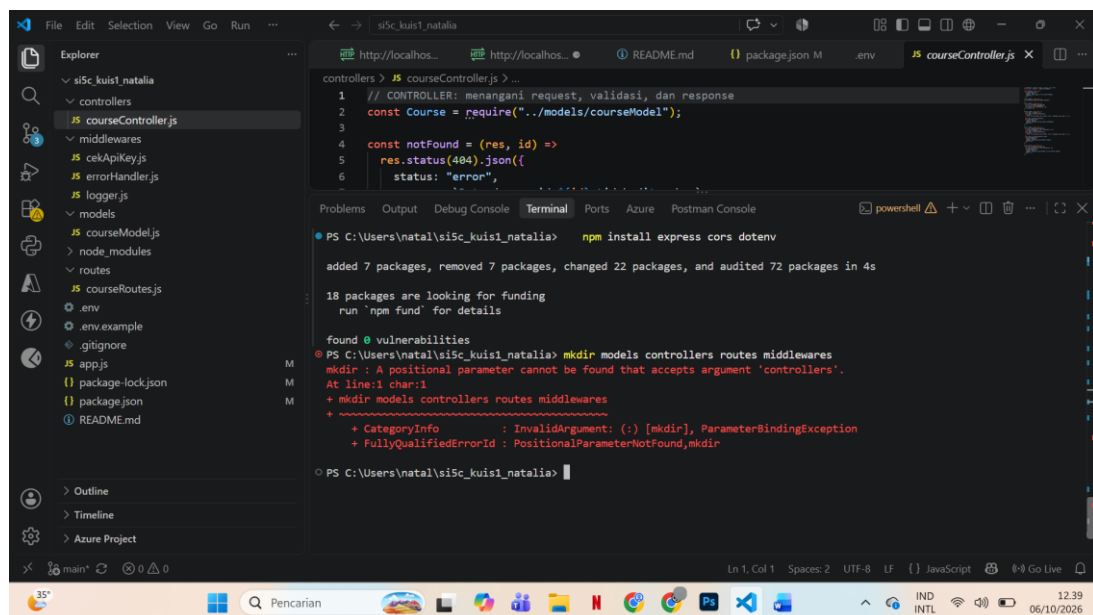
```
PS C:\Users\natal\si5c_kuis1_natalia> npm run dev
PS C:\Users\natal\si5c_kuis1_natalia> npm install express cors dotenv

added 7 packages, removed 7 packages, changed 22 packages, and audited 72 packages in 4s

18 packages are looking for funding
  run 'npm fund' for details

found 0 vulnerabilities
PS C:\Users\natal\si5c_kuis1_natalia>
```

B. Membuat folder Models, Controllers, Routes, dan middleware (Sudah terbuat di terminal sebelumnya, diketik ulang, sebagai bukti)



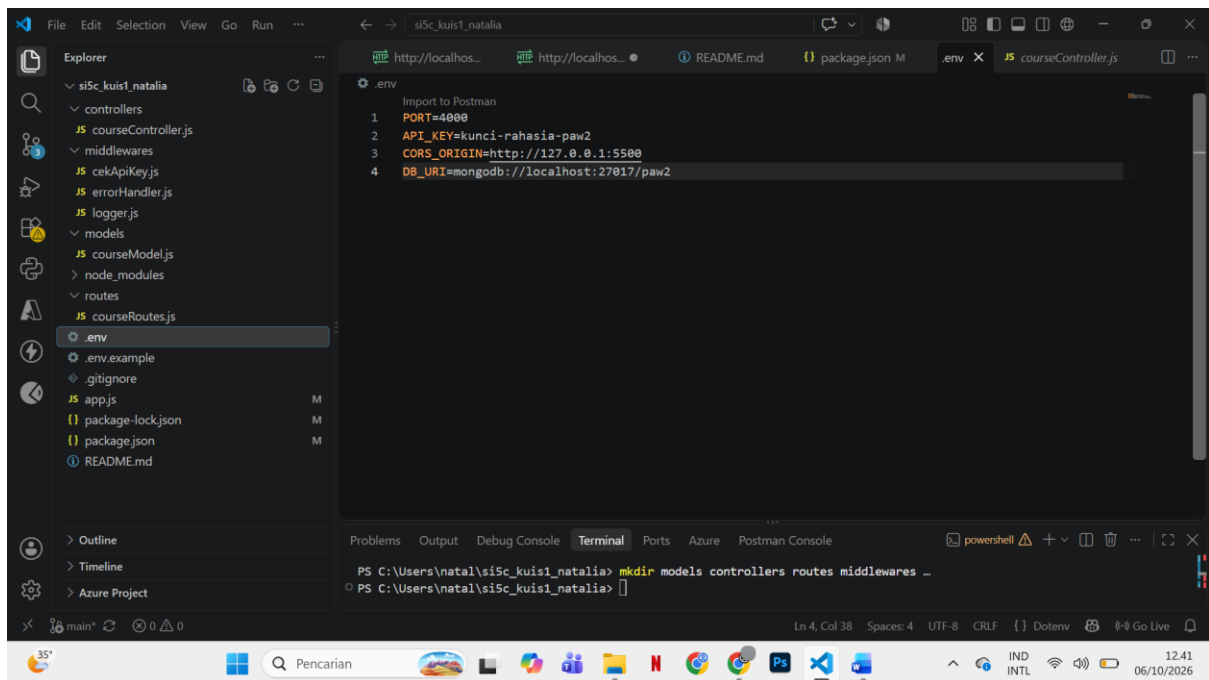
```
PS C:\Users\natal\si5c_kuis1_natalia> npm install express cors dotenv

added 7 packages, removed 7 packages, changed 22 packages, and audited 72 packages in 4s

18 packages are looking for funding
  run 'npm fund' for details

found 0 vulnerabilities
PS C:\Users\natal\si5c_kuis1_natalia> mkdir models controllers routes middlewares
mkdir : A positional parameter cannot be found that accepts argument 'controllers'.
At line:1 char:1
+ mkdir models controllers routes middlewares
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [mkdir], ParameterBindingException
+ FullyQualifiedErrorId : PositionalParameterNotFound,mkdir
PS C:\Users\natal\si5c_kuis1_natalia>
```

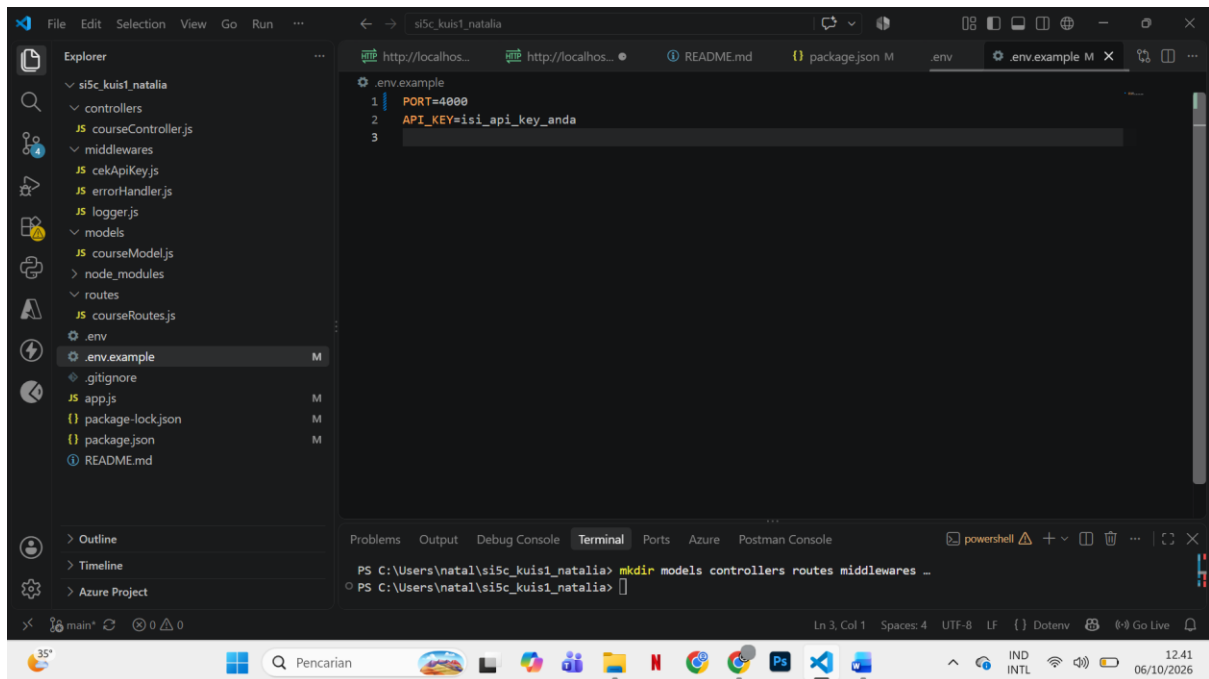
C. Mengatur .env, .env.example, dan .gitignore



This screenshot shows the Visual Studio Code interface with the `.env` file open. The file contains the following configuration:

```
1 PORT=4000
2 API_KEY=kunci-rahasia-paw2
3 CORS_ORIGIN=http://127.0.0.1:5500
4 DB_URI=mongodb://localhost:27017/paw2
```

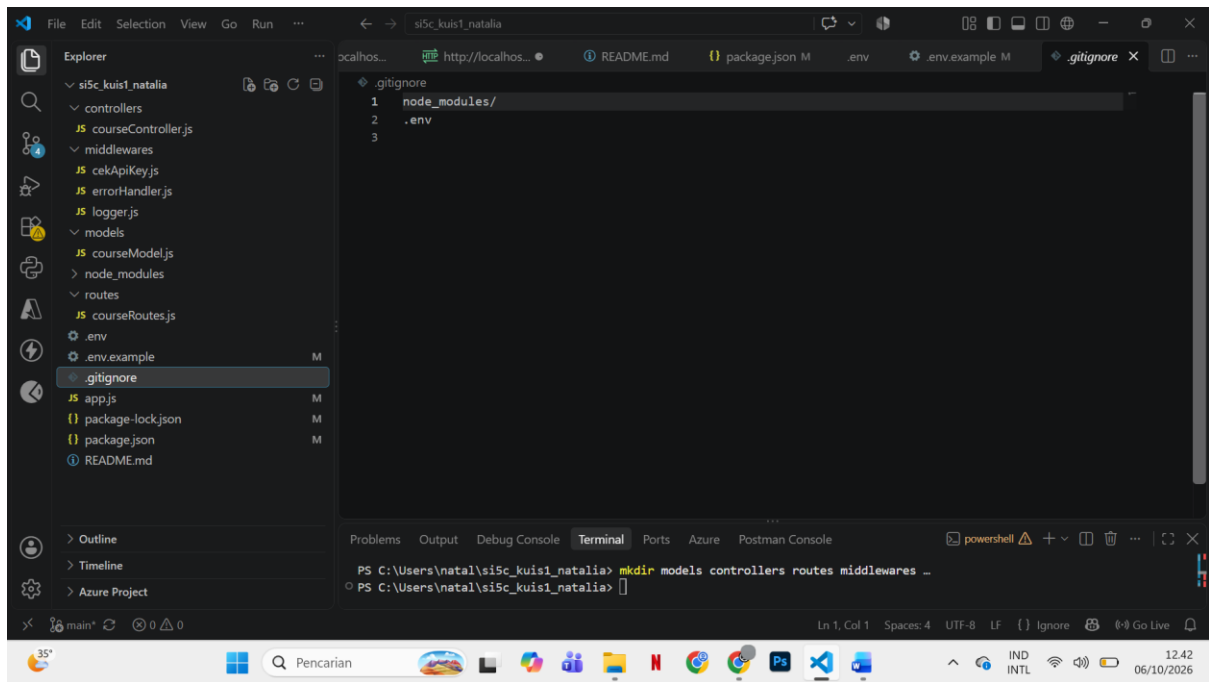
The Explorer sidebar on the left shows the project structure, with `.env` selected. The terminal at the bottom shows the command `mkdir models controllers routes middlewares` being executed.



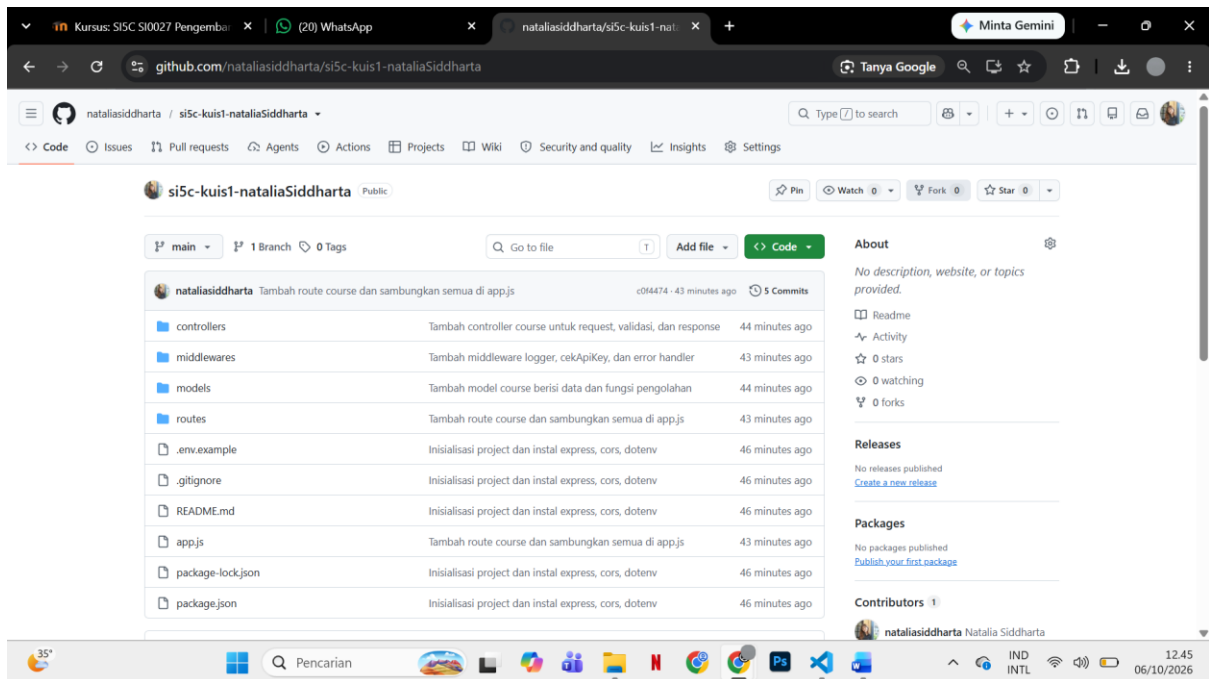
This screenshot shows the Visual Studio Code interface with the `.env.example` file open. The file contains the following configuration:

```
1 PORT=4000
2 API_KEY=isi_api_key_anda
3
```

The Explorer sidebar on the left shows the project structure, with `.env.example` selected. The terminal at the bottom shows the command `mkdir models controllers routes middlewares` being executed.



D. Membuat repository si5c-kuis1-natalia pada GitHub (sebelumnya masih kosong, di screenshot setelah sudah push)



E. PUSH FILE KE REPO

The screenshot shows the VS Code interface with a REST client request to `http://localhost:4000/courses/99`. The response is a 404 Not Found error with the message: `{ "status": "error", "message": "Data dengan id 99 tidak ditemukan", "data": null }`. The terminal shows the initial setup of the project:

```
PS C:\Users\natal\si5c_kuis1_natalia> npm run dev
Server berjalan di http://localhost:4000
PS C:\Users\natal\si5c_kuis1_natalia>
PS C:\Users\natal\si5c_kuis1_natalia> git init
Initialized empty Git repository in C:\Users\natal\si5c_kuis1_natalia/.git/
PS C:\Users\natal\si5c_kuis1_natalia> git branch -M main
PS C:\Users\natal\si5c_kuis1_natalia> git add package-lock.json package-lock.json .gitignore .env.example README.md
warning: in the working copy of '.env.example', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
```

The screenshot shows the terminal output for pushing the code to a GitHub repository:

```
PS C:\Users\natal\si5c_kuis1_natalia> git remote add origin https://github.com/nataliasiddharta/si5c-kuis1-nataliaSiddhar
ta.git
PS C:\Users\natal\si5c_kuis1_natalia> git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (7/7), 8.47 KiB | 1.06 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/nataliasiddharta/si5c-kuis1-nataliaSiddharta.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\natal\si5c_kuis1_natalia> git add models
warning: in the working copy of 'models/courseModel.js', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\natal\si5c_kuis1_natalia> git commit -m "Tambah model course berisi data dan fungsi pengolahan"
[main 9db5b0c] Tambah model course berisi data dan fungsi pengolahan
1 file changed, 68 insertions(+)
```

The screenshot shows the VS Code interface with the Explorer view on the left displaying the file structure of a project named 'si5c_kuis1_natalia'. The file structure includes folders for 'controllers', 'middlewares', 'models', 'routes', and 'node_modules', along with files like '.env', '.gitignore', 'package-lock.json', 'package.json', 'README.md', 'app.js', 'courseController.js', 'courseRoutes.js', 'cekApiKey.js', 'errorHandler.js', and 'logger.js'. The Terminal view on the right shows the following commands and output:

```
PS C:\Users\natal\si5c_kuis1_natalia> git push
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 1.10 KiB | 1.10 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/nataliasiddharta/si5c-kuis1-nataliaSiddharta.git
71be951..9db5b0c main -> main
PS C:\Users\natal\si5c_kuis1_natalia> git add controllers
warning: in the working copy of 'controllers/courseController.js', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\natal\si5c_kuis1_natalia> git commit -m "Tambah controller course untuk request, validasi, dan response"
[main 5ab03f0] Tambah controller course untuk request, validasi, dan response
1 file changed, 60 insertions(+)
create mode 100644 controllers/courseController.js
PS C:\Users\natal\si5c_kuis1_natalia> git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 862 bytes | 431.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/nataliasiddharta/si5c-kuis1-nataliaSiddharta.git
9db5b0c..5ab03f0 main -> main
PS C:\Users\natal\si5c_kuis1_natalia> git add middlewares
warning: in the working copy of 'middlewares/cekApiKey.js', LF will be replaced by CRLF the next time Git touches it
```

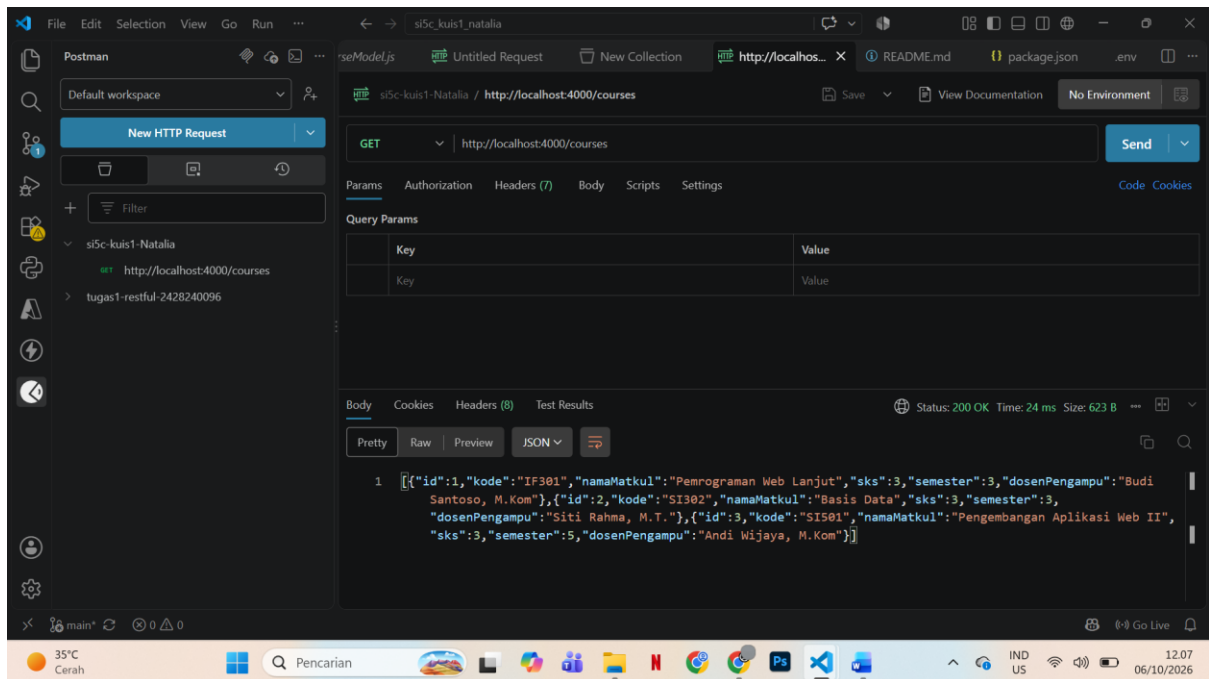
The screenshot shows the VS Code interface with the Explorer view on the left displaying the file structure of a project named 'si5c_kuis1_natalia'. The file structure includes folders for 'controllers', 'middlewares', 'models', 'routes', and 'node_modules', along with files like '.env', '.gitignore', 'package-lock.json', 'package.json', 'README.md', 'app.js', 'courseController.js', 'courseRoutes.js', 'cekApiKey.js', 'errorHandler.js', and 'logger.js'. The Terminal view on the right shows the following commands and output:

```
PS C:\Users\natal\si5c_kuis1_natalia> git add middlewares
warning: in the working copy of 'middlewares/logger.js', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\natal\si5c_kuis1_natalia> git commit -m "Tambah middleware logger, cekApiKey, dan error handler"
[main 48dc474] Tambah middleware logger, cekApiKey, dan error handler
3 files changed, 36 insertions(+)
create mode 100644 middlewares/cekApiKey.js
create mode 100644 middlewares/errorHandler.js
create mode 100644 middlewares/logger.js
PS C:\Users\natal\si5c_kuis1_natalia> git add routes app.js
warning: in the working copy of 'app.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'routes/courseRoutes.js', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\natal\si5c_kuis1_natalia> git commit -m "Tambah route course dan sambungkan semua di app.js"
[main c0f4474] Tambah route course dan sambungkan semua di app.js
2 files changed, 54 insertions(+)
create mode 100644 app.js
create mode 100644 routes/courseRoutes.js
PS C:\Users\natal\si5c_kuis1_natalia> git push
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 8 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (11/11), 2.07 KiB | 705.00 KiB/s, done.
Total 11 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/nataliasiddharta/si5c-kuis1-nataliaSiddharta.git
5ab03f0..c0f4474 main -> main
PS C:\Users\natal\si5c_kuis1_natalia> npm run dev
```

Di commit dan push perbagian dengan nama commit yang bermakna

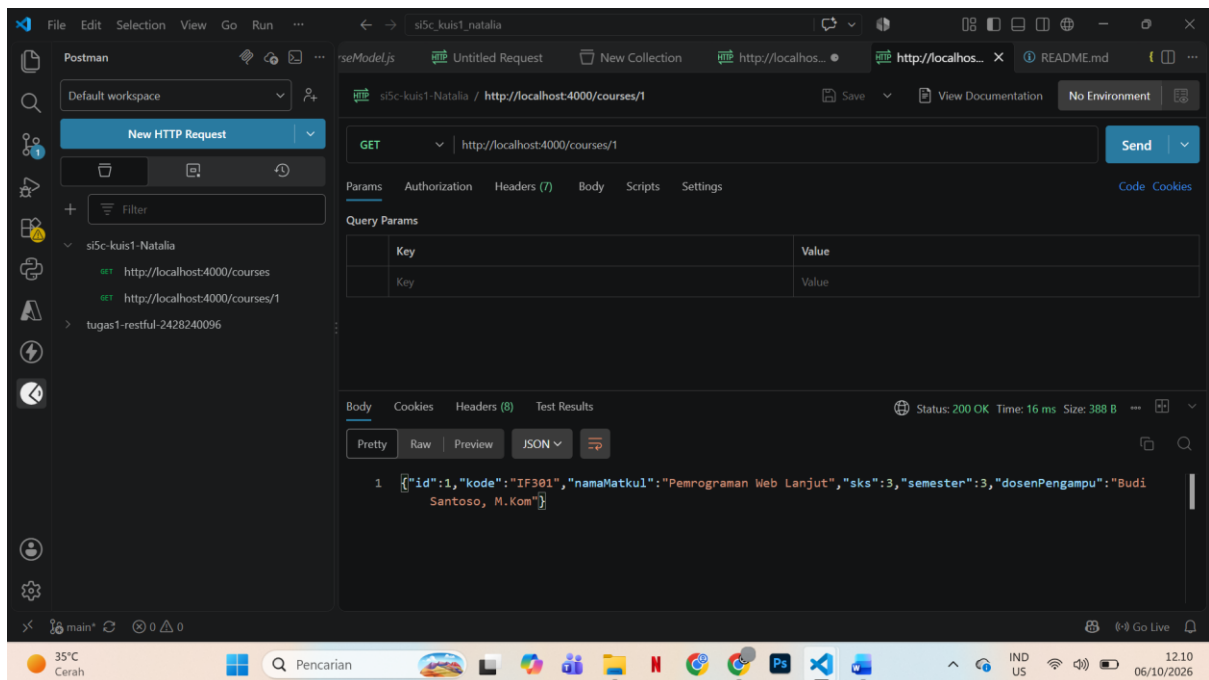
F. Pengujian ENDPOINT

1. GET semua data



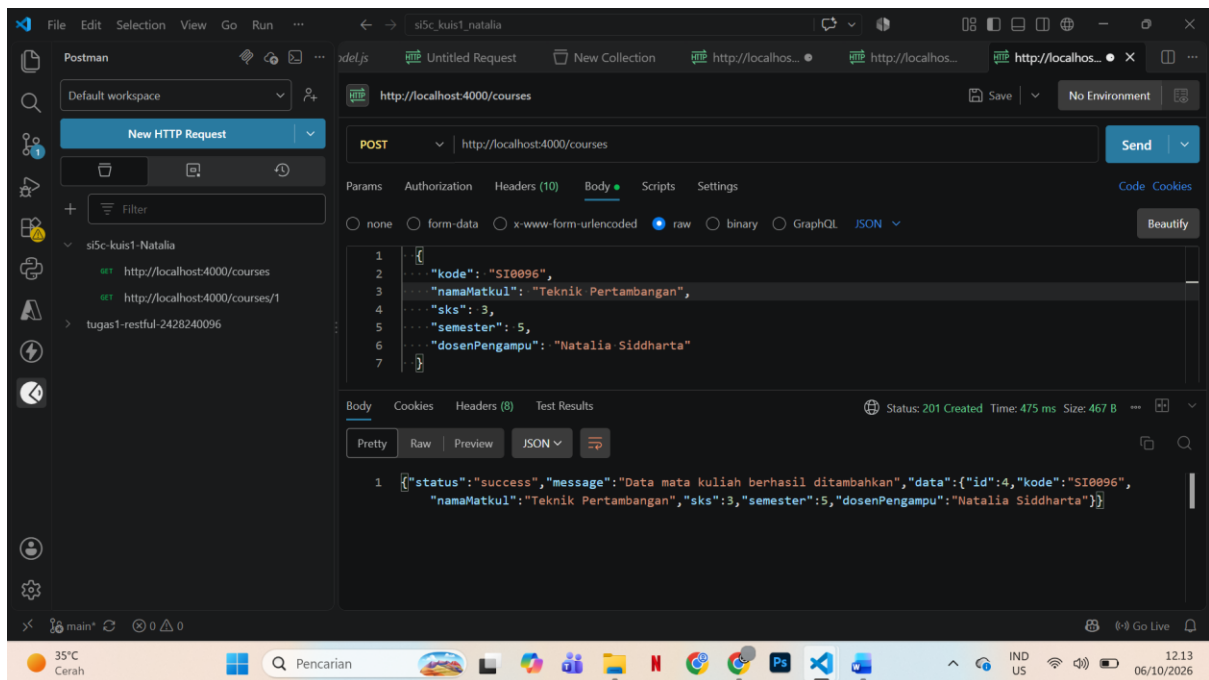
- Hasil: status **200 OK** dan array berisi 3 mata kuliah.
- Artinya: membuktikan data bisa dibaca dan tidak butuh API key.

2. Get Satu data berdasarkan ID



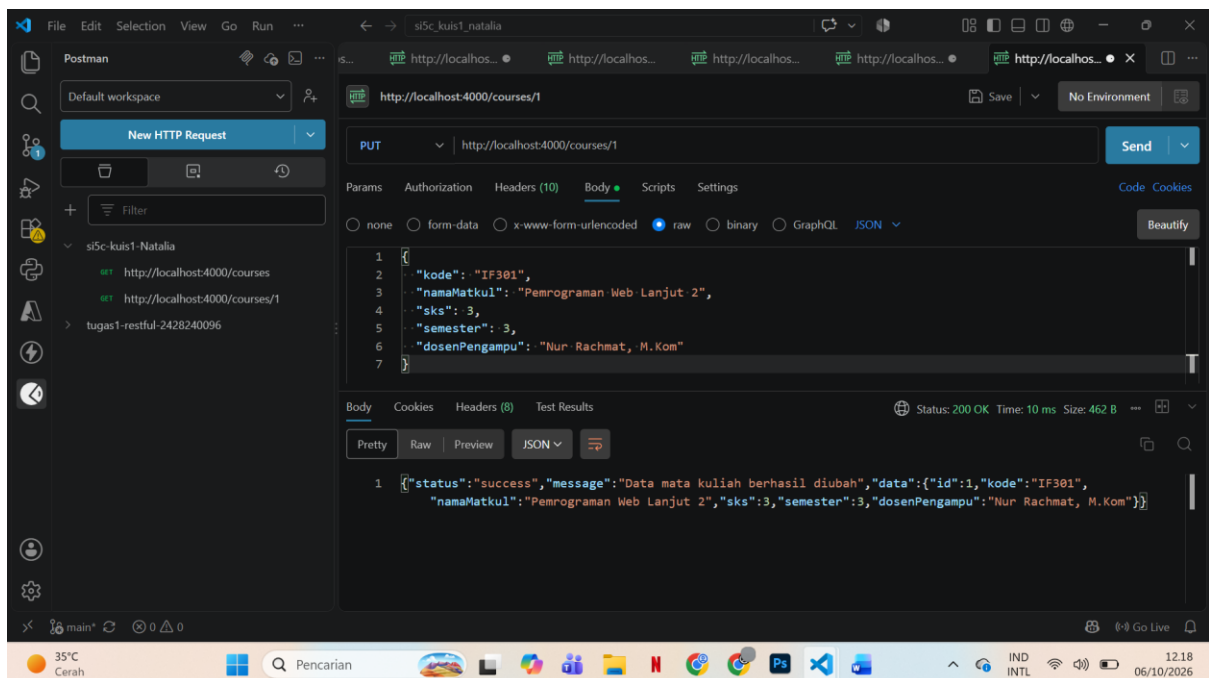
- Hasil: **200 OK** dan satu objek (kode IF301).
- Artinya: membuktikan pencarian per ID bekerja.

3. POST tambah data



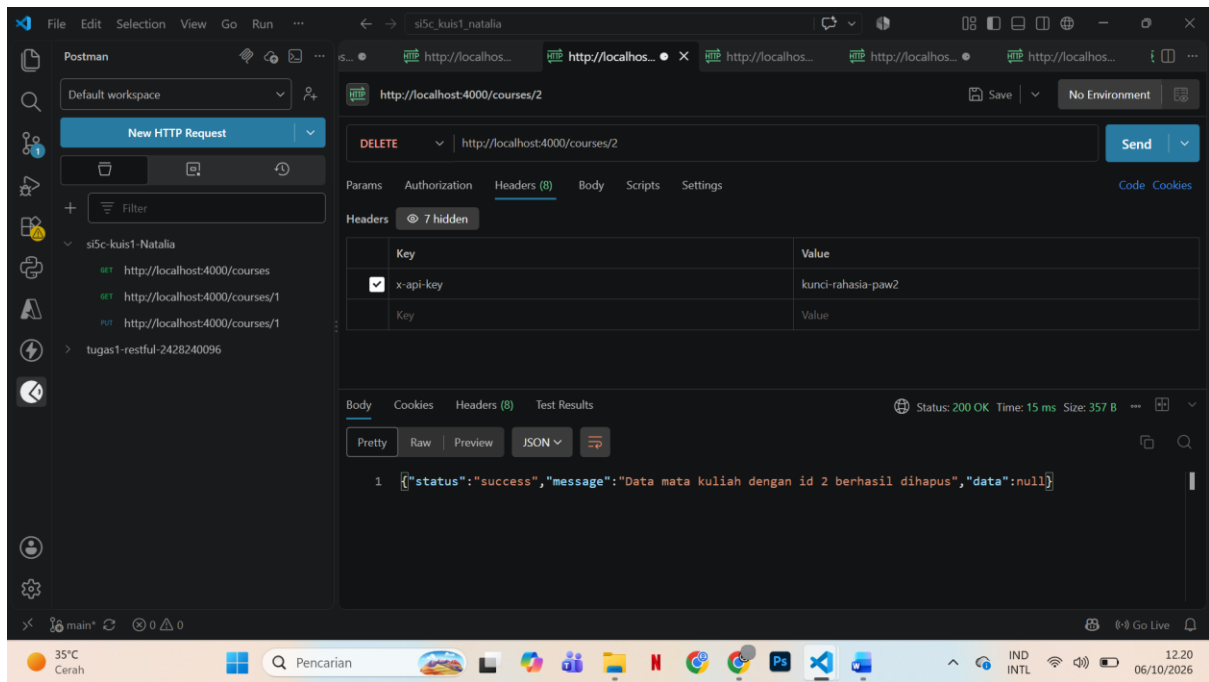
- Hasil: **201 Created**, dan data berisi id: 4.
- Artinya: membuktikan data baru tersimpan lewat route yang dilindungi API key.

4. PUT ubah data



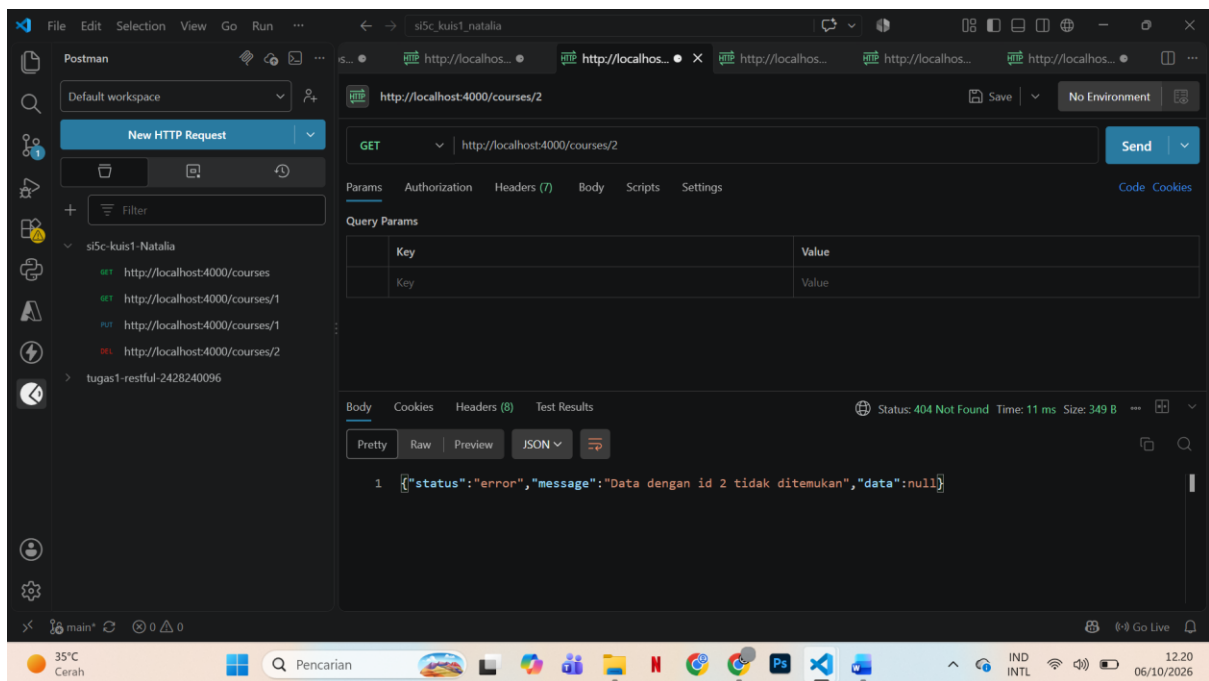
- Hasil: **200 OK** dan pesan "Data mata kuliah berhasil diubah".
- Artinya: PUT mengganti seluruh data lama dengan data baru, sedangkan ID tetap.

5. DELETE hapus data

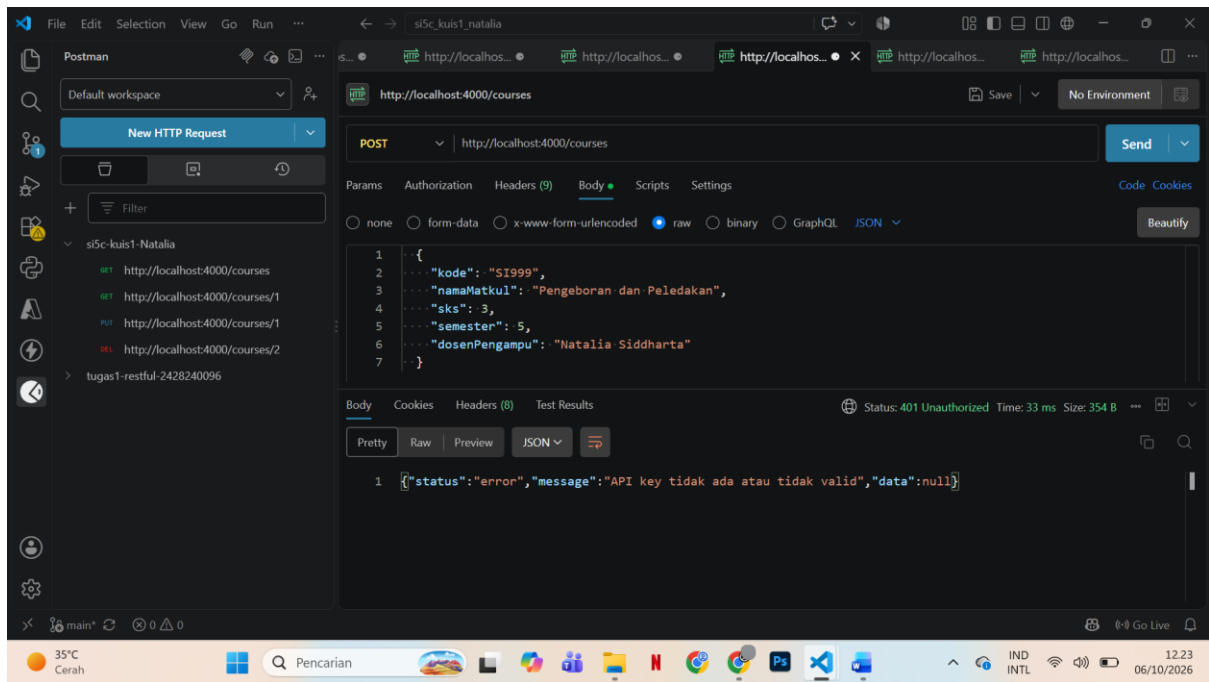


Hasil: **200 OK** dan pesan berhasil dihapus.

Bukti Sudah terhapus :



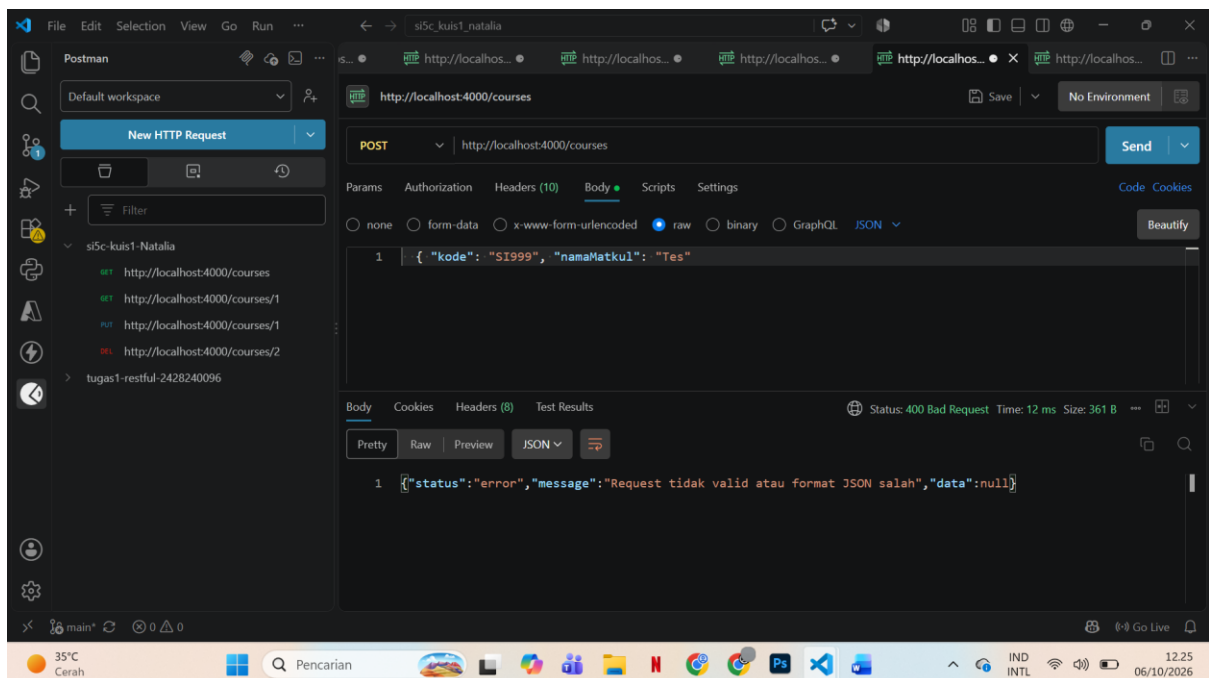
6. POST tanpa API Key



- Hasil: **401 Unauthorized**.
- Artinya: middleware cekApiKey memblokir request sebelum sampai ke controller.

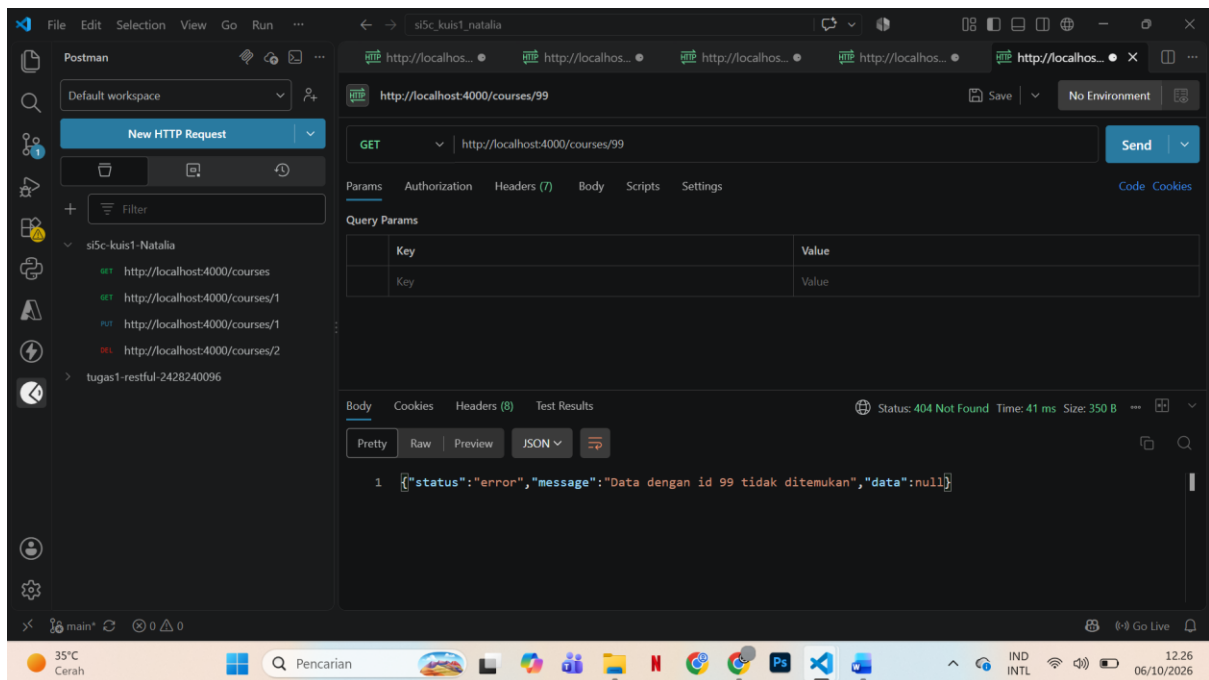
8. JSON rusak

- POST dengan header API key yang benar, body raw JSON yang sengaja salah



- Hasil: **400 Bad Request**, pesan "Request tidak valid atau format JSON salah".
- Artinya: error terpusat menangkap JSON yang gagal dibaca, sehingga server tidak crash dan tetap membalas dalam format JSON.

9. ID tidak ditemukan



- Hasil: **404 Not Found**, pesan "Data dengan id 99 tidak ditemukan".
- Artinya: model tidak menemukan data, lalu controller membalas 404.